TO:	Dr. Gergely Záruba, Chair, Department of Computer Science and Engineering
FROM:	Meetkumar Saspara
DATE:	March 22, 2026
SUBJECT:	Proposal to Integrate Docker and Containerization into the CS Curriculum

QUALIFICATION

I am a junior pursuing a B.S. in Computer Science at UNT with a GPA above 3.6 and an expected graduation of May 2027. I have completed coursework in software engineering, operating systems, and compiler design, and I have professional experience as a Software Engineering Intern at Yudiz Solutions. Through my internship and ongoing job search, I have seen firsthand that employers treat Docker as a baseline skill for software engineers—yet none of my UNT courses have covered it. This gap, along with the research presented below, motivates this proposal.

INTRODUCTION

This proposal seeks approval to integrate Docker and containerization tools into three existing upper-division courses: Software Engineering (CSCE 3530), Operating Systems (CSCE 3600), and senior Capstone projects. Rather than adding a new course—which would increase credit hours and student costs—the improvement embeds Docker into current course structures as a standard development tool. The proposal draws on an interview with a senior software engineer, a Qualtrics survey of 28 UNT CS students, and five secondary sources spanning industry data, educational research, and market analysis.

CURRENT SITUATION/PROBLEM

UNT's CS curriculum does not formally cover Docker or containerization in any required course. Students can earn their entire degree without ever writing a Docker file or managing a container. This is a problem because containerization is no longer optional in the industry. Docker's 2025 report shows that container usage among IT professionals has reached 92% (Docker, 2025), Docker appears in over 60% of Kubernetes-related job listings (Kube Careers, 2024), and the global container market is projected to grow from $5.85 billion to $31.5 billion by 2030 (Grand View Research, 2024).

UNT students confirm this gap. In my survey, 18 of 28 respondents said Docker was never covered in their courses, and of the 17 who had used Docker, none credited a UNT class—they learned through personal projects, internships, or self-study (Saspara, 2026). Meanwhile, 82% of respondents agreed that Docker should be formally taught. Jay Findoliya, a Senior Software Engineer I interviewed, described a common scenario in which a new graduate spent three days setting up a local development environment that Docker would have resolved in minutes. He estimated that developers without Docker experience need two to three months of on-the-job training before they can contribute independently (J. Findoliya, personal communication, February 8, 2026).

PROJECT PLAN

The integration follows the same model used when Git was adopted across CS courses—not as a standalone class but as a tool embedded in existing coursework. Research from Sánchez-Cifo et al. (2024) confirms that introducing DevOps tools within existing CS courses produces positive student outcomes, and Sorrel (2019) demonstrates that Docker simplifies classroom infrastructure rather than complicating it.

Target Courses

- **CSCE 3530 (Software Engineering):** Require group projects to use Docker Compose, eliminating the "works on my machine" problem.
- **CSCE 3600 (Operating Systems):** Use containers to demonstrate process isolation and resource management concepts hands-on.
- **Senior Capstone:** Require teams to deploy final deliverables in a containerized environment with a working Docker file.

Cost and Feasibility

Docker Desktop is free for educational use, runs on any modern laptop, and requires no new lab hardware. The primary investment is a two-day faculty workshop before the pilot semester. Since the integration modifies existing assignments rather than creating new courses, no curriculum committee approval for new course numbers is needed. Faculty workload is a valid concern, but the survey and interview data show that students are arriving at internships underprepared, and companies are spending months on training that the university could prevent.

PROJECT BENEFITS

For students, this change means graduating with documented experience in a tool that appears in the majority of software engineering postings. Professionals with containerization skills earn an average of $158,134 globally (Kube Careers, 2024), and 82% of surveyed students already want this content added. For the department, Docker integration signals alignment with industry standards and reduces a common headache in group-project courses: environment inconsistency. For employers, UNT graduates would arrive with skills that currently take months of on-the-job training to develop, reducing onboarding costs and strengthening UNT's reputation as a source of job-ready talent in the DFW area.

PROJECT TIMELINE

The table below outlines a two-semester implementation plan beginning in Summer 2026.

Phase	Task	Target Date
1	Present proposal to CSE leadership	May–Jun 2026
2	Faculty Docker workshop (two days)	Jul–Aug 2026
3	Pilot in one section of CSCE 3530	Fall 2026
4	Collect feedback; refine materials	Dec 2026–Jan 2027

5	Expand to CSCE 3600 and Capstone	Spring 2027
6	Evaluate outcomes; recommend adoption	May 2027

CONCLUSION

The data from every source I consulted points in the same direction: Docker is an industry standard, students feel unprepared, and integration is feasible within existing courses at minimal cost. This proposal asks the department to make a practical update that would improve student employability, strengthen the program's reputation, and reduce the gap between what we learn in the classroom and what employers expect on day one. Thank you for your time in reviewing this proposal. I welcome any questions or feedback and can be reached at MeetkumarSaspara@my.unt.edu.

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